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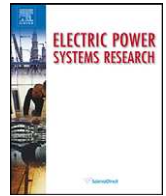
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Electricity demand load forecasting of the Hellenic power system using an ARMA model

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ABSTRACT

Effective modeling and forecasting requires the efficient use of the information contained in the available data so that essential data properties can be extracted and projected into the future. As far as electricity demand load forecasting is concerned time series analysis has the advantage of being statistically adaptive to data characteristics compared to econometric methods which quite often are subject to errors and uncertainties in model specification and knowledge of causal variables. This paper presents a new method for electricity demand load forecasting using the multi-model partitioning theory and compares its performance with three other well established time series analysis techniques namely Corrected Akaike Information Criterion (AICC), Akaike's Information Criterion (AIC) and Schwarz's Bayesian Information Criterion (BIC). The suitability of the proposed method is illustrated through an application to actual electricity demand load of the Hellenic power system, proving the reliability and the effectiveness of the method and making clear its usefulness in the studies that concern electricity consumption and electricity prices forecasts.

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1. Introduction

Load forecasting plays an important role in power system planning and operation. Basic operation functions such as unit commitment, economic dispatch, fuel scheduling and unit maintenance can be performed efficiently with an accurate forecast. Forecasting of electricity demand load is a recurrent, however not a routine, requirement in the management of utilities. Various degrees of sophistication exist in the available methods of forecasting, ranging from simple data extrapolation [1,2] and artificial intelligence techniques [3,4] to complex econometric models [5]. Recently new techniques for short-term load forecasting have been developed. In [6] a new algorithm for successful short-term forecasting is introduced by using a sample bispectrum in order to test whether the load data is Gaussian or not. In [7] a method for successful short-term load forecasting based on periodic time series analysis is proposed. Additionally the stationary properties of the estimated models are used in order to identify typical daily customer profiles of residential, business and industrial customers. The choice

of a specific approach depends upon considerations such as the required quality of forecasts, availability of input information, ease of application and cost of adoption. Usually difficulties in forecasting occur due to multiple seasonality (corresponding to weekly and yearly seasonality), calendar effect (such as weekends and holidays), high volatility, etc.

This study addresses the problem of modeling and forecasting the electricity demand loads of the Hellenic power system. Firstly an appropriate deseasonalization of the provided electricity demand load data covering the period from January 1st 2004 to December 31st 2005 is conducted. Then an Autoregressive Moving Average (ARMA) model is fitted (off-line) on this data using the Corrected Akaike Information Criterion (AICC). The developed model is shown to fit the data in a successful manner.

This model is used by four different estimation methods, a new method namely multi-model partitioning theory (MMPF), which is extensively presented in this paper, Corrected Akaike Information Criterion (AICC) [8], Akaike's Information Criterion (AIC) [9] and Schwarz's Bayesian Information Criterion (BIC) [10], in order to predict the electricity demand load for the period from January 1st 2006 to December 31st 2006. For every day in the test period an adaptive day-ahead prediction is adopted, meaning that instead of using a single model for the whole sample, for everyday in the

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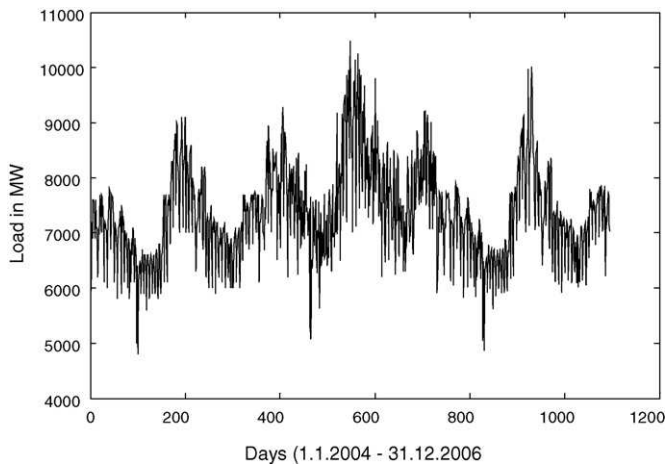


Fig. 1. Hellenic power market daily system-wide load from January 1st 2004 to December 31st 2006. The annual seasonality, the weekly seasonality and the increase in the load demand are clearly visible.

test period the best ARMA is adjusted to the previous 730 values of the deseasonalized data in order to obtain a forecasted value for that day. The developed model can be proved very useful in the studies of electric utilities' electrical engineers, which concern electricity consumption and electricity prices forecasts. It can be used as an alternative reliable tool in the very complex and important electricity demand load forecasting process.

2. Data preparation

The actual electricity demand load data used in this work has been provided by the Hellenic Public Power Corporation S.A. [11]. This data refer to a time series that contains the daily electricity demand load covering the period from January 1st 2004 to December 31st 2006. Figs. 1 and 2 clearly indicate the weekly and annual seasonality of the provided data. The first necessary step is to remove this feature. This is easily achieved by applying to the data a new technique analytically introduced by Nowicka-Zagrajek and Weron [12,13]. Furthermore the Modeling and Forecasting Electricity loads and prices (MFE) toolbox for Matlab® has been used in this study provided by Weron [14].

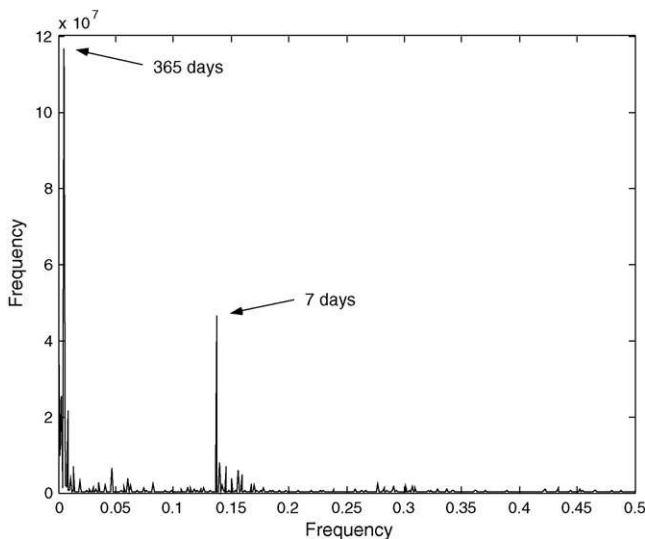


Fig. 2. Periodogram of the Hellenic power market daily system-wide load from January 1st 2004 to December 31st 2005. The annual and weekly frequencies are clearly visible.

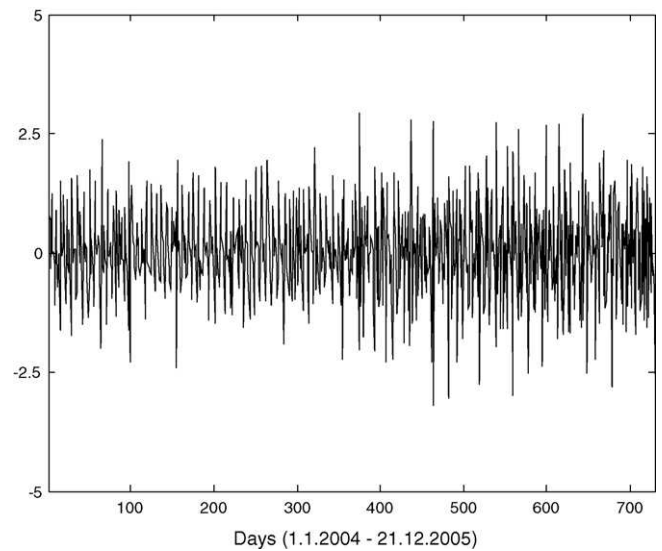


Fig. 3. Load returns after removal of the weekly and seasonal cycles.

2.1. Modeling with ARMA processes

The resulting time series (Figs. 3 and 4) show no apparent trend or seasonality. It can be considered as a realization of a stationary process. In addition to that, both the Auto Correlation Function (ACF) and Partial Auto Correlation Function (PACF) tend very fast to zero (Figs. 5 and 6) showing that the deseasonalized data returns can be modeled by an AR or an ARMA model.

An m -variate ARMA model of order (p, q) [ARMA (p, q)] for a stationary time series of vectors $\mathbf{y}$ observed at equally spaced instants $k = 1, 2, n$ is defined as:

$$\mathbf{y}_k = \sum_{i=1}^p \mathbf{A}_i \mathbf{y}_{k-i} + \sum_{j=1}^q \mathbf{B}_j \mathbf{v}_{k-j} + \mathbf{v}_k, \quad E[\mathbf{v}_k \mathbf{v}_k^T] = \mathbf{R} \quad (1)$$

where the m -dimensional vector $\mathbf{v}_k$ is uncorrelated random noise, not necessarily Gaussian, with zero-mean and covariance matrix $\mathbf{R}$, $\theta = (p, q)$ is the order of the predictor and $\mathbf{A}_1, \dots, \mathbf{A}_p, \mathbf{B}_1, \dots, \mathbf{B}_q$ are the $m \times m$ coefficient matrices of the Multivariate (MV) ARMA model. It is obvious that the problem is twofold. The first and probably the most important task is the successful determination of the

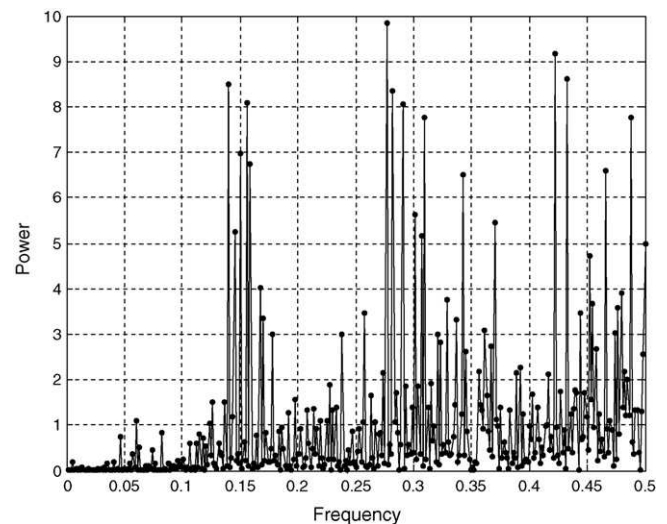


Fig. 4. Periodogram of the load returns after removal of the weekly and annual cycles. No dominating frequency can be observed.

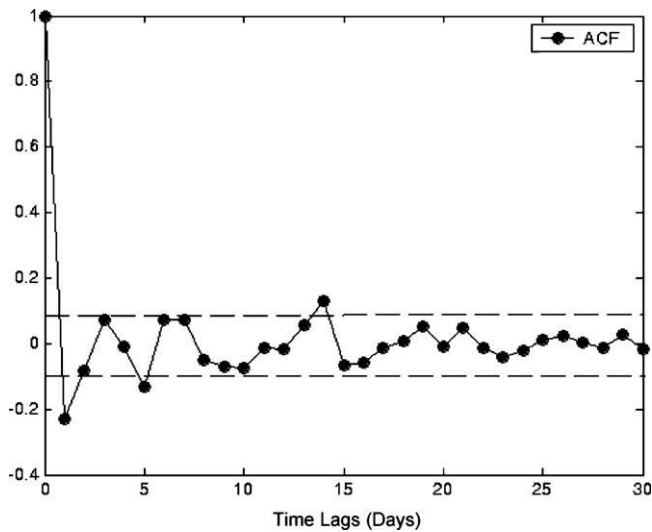


Fig. 5. The ACF for the mean corrected deseasonalized load returns. Dashed lines represent the bounds of $\pm 1.96/\sqrt{730}$, i.e., the 95% confidence intervals of Gaussian white noise.

predictor's order $\theta=(p, q)$ and the second task is the estimation of the predictor's matrix coefficients $\{\mathbf{A}_i, \mathbf{B}_j\}$.

Determining the order of the ARMA process is usually the most important part of the problem. Over the past years substantial literature has been produced for this problem and various different criteria, such as Akaike's, Rissanen's, Schwarz's, Wax's [9,10,15–17], have been proposed to implement the order selection process. Using the real data provided *off-line*, ARMA model order, $\theta=(p, q)$ identification and parameter estimation was accomplished by minimizing the Akaike's Corrected Information Criterion (AICC):

$$\text{AICC} = \log \left| \hat{\mathbf{R}}_{\theta} \right| + \frac{2(p+q+1)n}{n-p-q-2} \quad (2)$$

where $n=730$ is the sample size and $\theta(p, q)$ is the model order and $\hat{\mathbf{R}}$ is a maximum likelihood estimate of $\mathbf{R}$ under the assumption that ARMA (θ) is the correct model order [8].

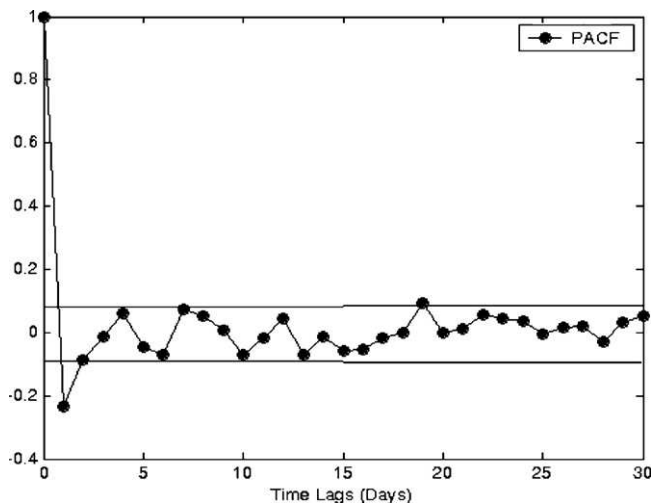


Fig. 6. The PACF for the mean deseasonalized load returns. Solid lines represent the bounds of $\pm 1.96/\sqrt{730}$, i.e., the 95% confidence intervals of Gaussian white noise.

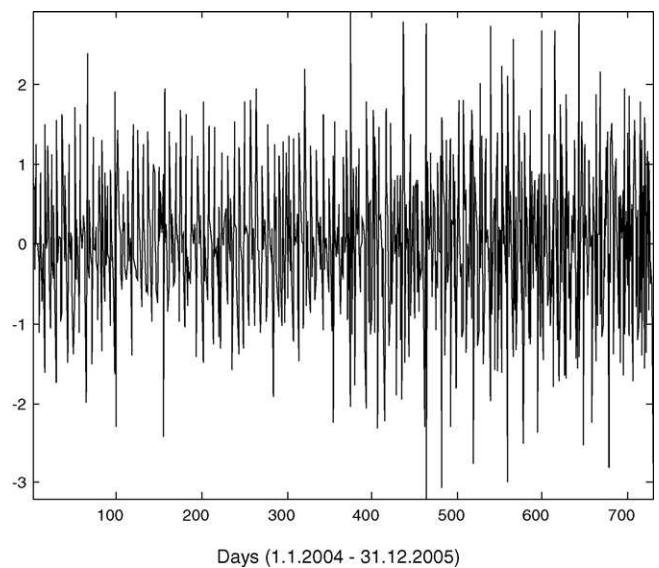


Fig. 7. ARMA (2, 6) behavior.

The optimization procedure led us to the following ARMA (2, 6) model with parameters:

$$\mathbf{y}_k = \sum_{i=1}^2 \mathbf{A}_i \mathbf{y}_{k-i} + \sum_{j=1}^6 \mathbf{B}_j \mathbf{v}_{k-j} + \mathbf{v}_k \quad (3)$$

$\mathbf{A}_1 = [-0.3153]$, $\mathbf{A}_2 = [0.0287]$, $\mathbf{B}_1 = [-0.1363]$, $\mathbf{B}_2 = [0.1985]$, $\mathbf{B}_3 = [0.1604]$, $\mathbf{B}_4 = [0.4057]$, $\mathbf{B}_5 = [0.4211]$, $\mathbf{B}_6 = [0.2417]$. The value of the AICC criterion obtained for this model was $\text{AICC} = 1821.362$.

The obtained ARMA (2, 6) model fits satisfactorily the deperseasonalized load returns (Fig. 7). The graph shows no indication of a non-zero or non-constant variance. In addition to that both ACF and PACF fall between the calculated bounds proving that there is no correlation in the series (Figs. 8 and 9). Moreover using the *Portmanteau test* [12] it was found that $Q = 18.21$ and $p\text{-value} = 0.8149$ for $\alpha = 5\%$. From Statistical tables it is found that $\chi^2_{1-\alpha} = 31.41$ for $\alpha = 0.05$ and since $Q < \chi^2_{1-\alpha}$ no reason to reject the fitted model.

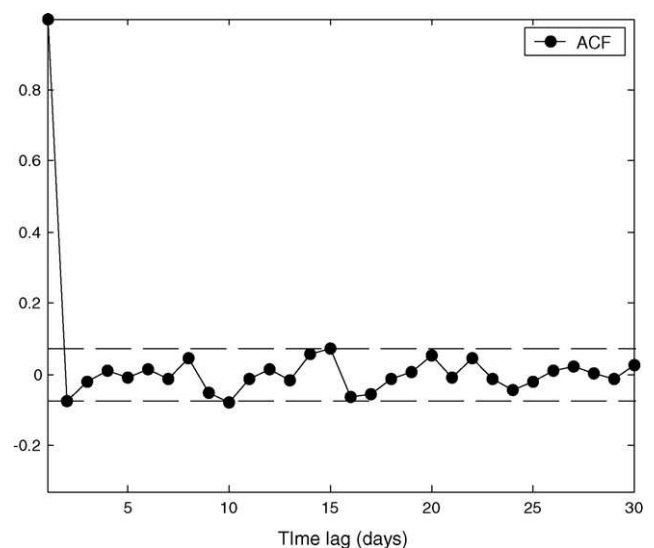


Fig. 8. The ACF for the obtained ARMA (2, 6) model. Dashed lines represent the bounds of $\pm 1.96/\sqrt{730}$, i.e., the 95% confidence intervals of Gaussian white noise.

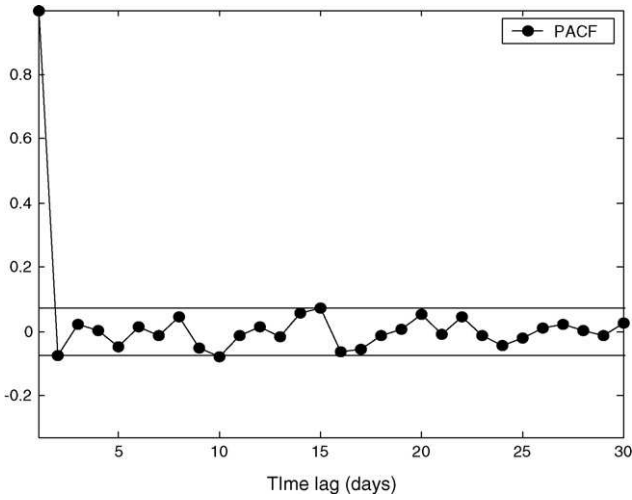


Fig. 9. The PACF for the obtained ARMA(2, 6) model. Solid lines represent the bounds of $\pm 1.96/\sqrt{730}$, i.e., 95% confidence intervals of Gaussian white noise.

2.2. Problem reformulation using multi-model partitioning filter (MMPF)

The model used in this study is a univariate model. The new algorithm for ARMA model order and parameter estimation proposed concerns the general multivariate case. The method is easily transformed to univariate by just substituting the matrix coefficients $\{\mathbf{A}_i, \mathbf{B}_j\}$ with real numbers, by setting dimensionality $m=1$ and re-writing (4)–(11) appropriately without the vector notation (bold style).

Assuming that the model order fitting the data is known and is equal to $\theta=(p, q)$, (1) can be written in standard state-space form as:

$$\mathbf{x}(k+1) = \mathbf{x}(k) \quad (4)$$

$$\mathbf{y}(k) = \mathbf{H}(k)\mathbf{x}(k) + \mathbf{v}(k) \quad (5)$$

where $\mathbf{x}(k)$ is an $m^2(p+q) \times 1$ vector made up from the coefficients of the matrices $\{\mathbf{A}_1, \dots, \mathbf{A}_p, \mathbf{B}_1, \dots, \mathbf{B}_q\}$, and $\mathbf{H}(k)$ is an $m \times m^2(p+q)$ observation history matrix of the process $\{\mathbf{y}(k)\}$ up to time $k-(p+q)$.

Assuming that the general forms of the matrices $\mathbf{A}_p$ and $\mathbf{B}_q$ are as follows:

$$\begin{bmatrix} a_{11}^p & \dots & a_{1m}^p \\ \vdots & \ddots & \vdots \\ a_{m1}^p & \dots & a_{mm}^p \end{bmatrix} \quad (6)$$

$$\begin{bmatrix} b_{11}^q & \dots & b_{1m}^q \\ \vdots & \ddots & \vdots \\ b_{m1}^q & \dots & b_{mm}^q \end{bmatrix} \quad (7)$$

$$\mathbf{x}(k) \triangleq [\alpha_{11}^1 \alpha_{21}^1 \dots \alpha_{m1}^1 : \alpha_{12}^1 \alpha_{22}^1 \dots \alpha_{m2}^1 : \dots : \alpha_{1m}^p \alpha_{2m}^p \dots \alpha_{mm}^p : b_{11}^1 b_{21}^1 \dots b_{m1}^1 : b_{12}^1 b_{22}^1 \dots b_{m2}^1 : \dots : b_{1m}^q b_{2m}^q \dots b_{mm}^q]^T \quad (8)$$

$$\mathbf{H}(k) \triangleq [y_1(k-1) \mathbf{I} \dots y_m(k-1) \mathbf{I} : \dots : y_1(k-p) \mathbf{I} \dots y_m(k-p) \mathbf{I} : v_1(k-1) \mathbf{I} \dots v_m(k-1) \mathbf{I} : \dots : v_1(k-q) \mathbf{I} \dots v_m(k-q) \mathbf{I}] \quad (9)$$

where $\mathbf{I}$ is the $m \times m$ identity matrix and $\theta=(p, q)$, is the model order.

In case that the system model and its statistics were completely known, the Kalman filter (KF) in its various forms would be the optimal estimator in the minimum variance sense. Moreover in case

that the prediction coefficients are subject to random perturbations (4) becomes:

$$\mathbf{x}(k+1) = \mathbf{x}(k) + \mathbf{w}(k) \quad (10)$$

where $\mathbf{v}(k)$, $\mathbf{w}(k)$ are independent, zero-mean, white processes, not necessarily Gaussian.

The form of $\mathbf{w}(k)$ is:

$$\mathbf{w}(k) \triangleq [w_{11}^1 w_{21}^1 \dots w_{m1}^1 : w_{12}^1 w_{22}^1 \dots w_{m2}^1 : \dots : w_{1m}^p w_{2m}^p \dots w_{mm}^p : w_{11}^1 w_{21}^1 \dots w_{m1}^1 : w_{12}^1 w_{22}^1 \dots w_{m2}^1 : \dots : w_{1m}^q w_{2m}^q \dots w_{mm}^q]^T \quad (11)$$

A complete system description requires the value assignments of the variances of the random processes $\mathbf{w}(k)$ and $\mathbf{v}(k)$. Adopting the usual assumption that $\mathbf{w}(k)$ and $\mathbf{v}(k)$ at least wide sense stationary processes, hence their variances, $\mathbf{Q}$ and $\mathbf{R}$ respectively are time invariant. To obtain these values is not always trivial. If $\mathbf{Q}$ and $\mathbf{R}$ are not known they can be estimated by using a method such as the one described in [18]. In the case of coefficients constant in time, or slowly varying, $\mathbf{Q}$ is assumed to be zero.

It is also necessary to assume an a priori mean and variance for each $\{\mathbf{A}_i, \mathbf{B}_j\}$. The a priori mean of the $\mathbf{A}_i(0)$'s and $\mathbf{B}_j(0)$'s can be set to zero if no knowledge about their values is available before any measurements are taken (the most likely case). On the other hand the usual choice of the initial variance of the $\mathbf{A}_i$'s and $\mathbf{B}_j$'s, denoted by $\mathbf{P}_0$ is $\mathbf{P}_0 = n\mathbf{I}$, where n is a large integer.

Considering the case where the system model is not completely known the adaptive multi-model partitioning filter (MMPF) is one of the most widely used approaches for similar problems. This approach was introduced by Lainiotis [19–21] and summarizes the parametric model uncertainty into an unknown, finite dimensional parameter vector whose values are assumed to lie within a known set of finite cardinality. A non-exhaustive list of the reformulation, extension and application of the MMPF approach as well as its application to a variety of problems can be found in [22–30]. In the problem studied in this paper it is assumed that the model uncertainty is the lack of knowledge of the model order θ . It is also assumed that the model order θ lies within a known set of finite cardinality: $1 \leq \theta \leq M$, where $\theta=(p, q)$, is the model order.

The MMPF operates on the following discrete model:

$$\mathbf{x}(k+1) = \mathbf{F}(k+1, k/\theta) \mathbf{x}(k) + \mathbf{w}(k) \quad (12)$$

$$\mathbf{y}(k) = \mathbf{H}(k/\theta) \mathbf{x}(k) + \mathbf{v}(k) \quad (13)$$

where $\theta=(p, q)$ is the unknown parameter, i.e., the model order. A block diagram of the MMPF is presented in Fig. 10.

In the Gaussian case the optimal MMSE estimate of $\mathbf{x}(k)$ is given by:

$$\hat{\mathbf{x}}(k/k) = \sum_{j=1}^M \hat{\mathbf{x}}(k/k; \theta_j) p(\theta_j/k) \quad (14)$$

finite set of models is designed, each matching one value of the parameter vector. In the case that the prior probabilities $p(\theta_j/k)$ for each model are already known, these are assigned to each model. In the absence of any prior knowledge, these are set to $p(\theta_j/k) = 1/M$ where M is the cardinality of the model set.

A bank of conventional elemental filters (non-adaptive, e.g. Kalman) is then applied, one for each model, which can be run in parallel. At each iteration, the MMPF selects the model which corresponds to the maximum posteriori probability as the correct one. This probability tends to 1, while the others tend to 0. The overall optimal estimate can be taken either to be the individual estimate of the elemental filter exhibiting the highest posterior probability, called the maximum posteriori (MAP) estimate [27],

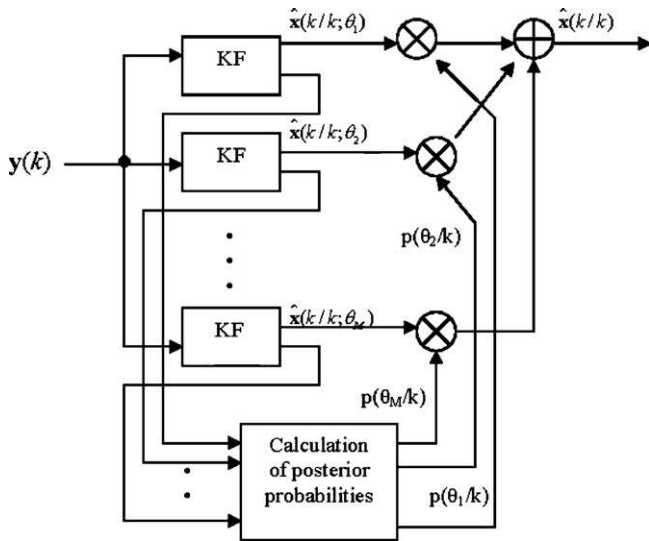


Fig. 10. MMPF block diagram.

or the weighted average of the estimates produced by the elemental filters, as described in (14), and is the case used in this paper. The weights are determined by the posterior probability that each model in the model set is in fact the true model. The posterior probabilities are calculated on-line in a recursive manner as follows:

$$p(\theta_j/k) = \frac{L(k/k; \theta_j)}{\sum_{j=1}^M L(k/k; \theta_j) p(\theta_j/k-1)} p(\theta_j/k-1) \quad (15)$$

$$L(k/k; \theta_j) = |\mathbf{P}_{\tilde{\mathbf{y}}}(k/k-1; \theta_j)|^{-1/2} \cdot \exp \left[-\frac{1}{2} \tilde{\mathbf{y}}^T \times (k/k-1; \theta_j) \mathbf{P}_{\tilde{\mathbf{y}}}^{-1}(k/k-1; \theta_j) \tilde{\mathbf{y}}(k/k-1; \theta_j) \right] \quad (16)$$

where the innovation process

$$\tilde{\mathbf{y}}(k/k-1; \theta_j) = \mathbf{y}(k) - \mathbf{H}(k; \theta_j) \hat{\mathbf{x}}(k/k-1; \theta_j) \quad (17)$$

is a zero-mean white process with covariance matrix

$$\mathbf{P}_{\tilde{\mathbf{y}}}(k/k-1; \theta_j) = \mathbf{H}(k; \theta_j) \mathbf{P}(k/k; \theta_j) \mathbf{H}^T(k; \theta_j) + \mathbf{R} \quad (18)$$

It must be mentioned that in (9)–(18) the value of $j=1, 2, \dots, M$. An important feature of the MMPF is that all the Kalman filters needed to be implemented can be independently realized. This enables to implement them in parallel, saving an enormous computational time [27]. Eqs. (14) and (15) refer to the current case where the sample space is naturally discrete. However in real world applications, θ 's probability density function (pdf) is continuous and an infinite number of Kalman filters have to be applied for the exact realization of the optimal estimator. The usual approximation considered to overcome this difficulty is to approximate θ 's pdf by a finite sum. Many discretization strategies have been proposed in the literature and some of them are presented in [31–32]. When the true parameter value lies outside the assumed sample space, the adaptive estimator converges to the value that in the sample space which is closer (i.e. minimizes the Kullback Information Measure) to the true value [33]. This means that the value of the unknown parameter cannot be exactly defined. The application of hybrid techniques that combine the MMPF with Genetic Algorithms are able to overcome this difficulty [24,34].

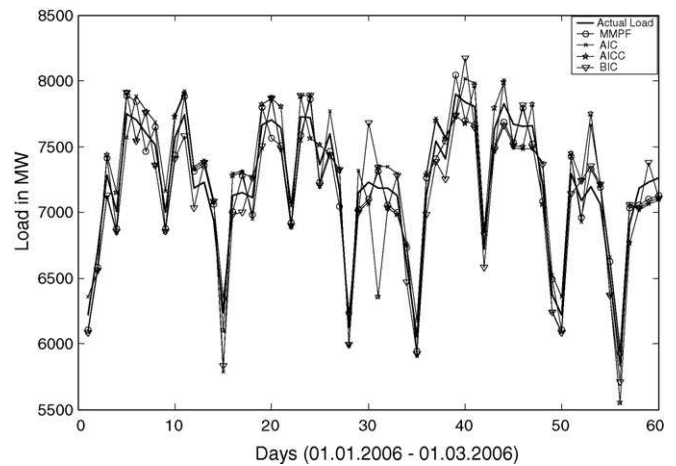


Fig. 11. Hellenic power market daily system-wide load from January 1st 2006 to March 1st 2006, compared with MMPF, AIC, AICC, BIC day-ahead forecasts.

3. Applications

3.1. Mid-range load forecast

The problem of ARMA modeling is much more difficult when an adaptive on-line procedure is required. The earlier mentioned criteria, Akaike's, Rissanen's, Schwarz's, and Wax's [9,10,15–17], are not always optimal and are also known to suffer from deficiencies; for example, Akaike's information criterion suffers from over fit [35]. Also their performance depends on the assumption that the data is Gaussian and upon asymptotic results. In addition to this, their applicability is justified only for large samples; furthermore, they are two pass methods, so they cannot be used in an on-line or adaptive fashion. At this point it should be stated firstly that AICC performs better than the classical AIC that is why it is used in this research and secondly the noise is considered to be Gaussian.

In Section 2 an ARMA model was fitted to the deseasonalized data covering the period from January 1st 2004 to December 31st 2005. This model will be used by four different estimation methods in order to predict the electric load for the period from January 1st 2006 to December 31st 2006. For every day in the test period an adaptive day-ahead prediction is adopted, meaning that instead of using a single model for the whole sample, for everyday in the test period the best ARMA (2, 6) is adjusted to the previous 730 values of the deseasonalized data in order to obtain a forecasted value for that day. A comparison amongst the following estimation criteria will be performed.

- Akaike's Corrected Information Criterion (AICC), see (2).
- Akaike Information Criterion (AIC).

$$\log(|\hat{\mathbf{R}}_{\theta}|) + \frac{2(p+q)}{n} \quad (19)$$

- Bayesian Information Criterion (BIC).

$$n \log(|\hat{\mathbf{R}}_{\theta}|) + (p+q) \log(n) \quad (20)$$

- MMPF.

The performance of the model used with four different estimation techniques can be seen in Figs. 11–13.

As it can be seen from Table 1 all of the estimation techniques present an acceptable performance. However MMPF has the smallest MAPE so it can be said that is the best amongst the four of them.

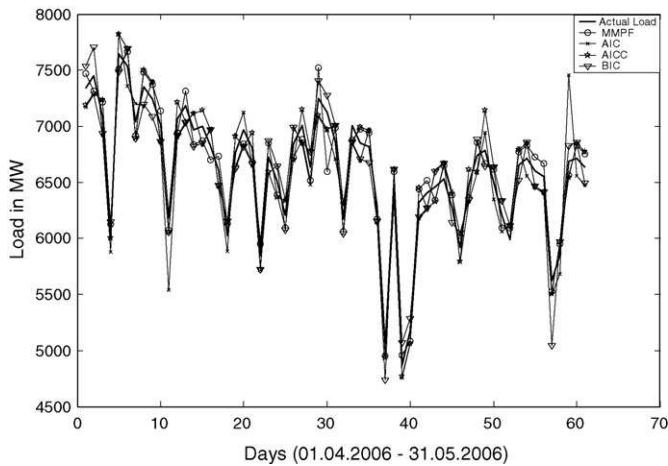


Fig. 12. Hellenic power market daily system-wide load from April 1st 2006 to May 31st 2006, compared with MMPF, AIC, AICC, BIC day-ahead forecasts.

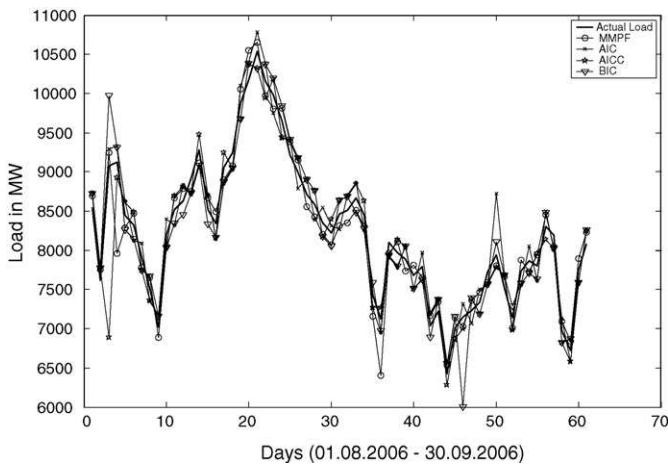


Fig. 13. Hellenic power market daily system-wide load from August 1st 2006 to September 30th 2006, compared with MMPF, AIC, AICC, BIC day-ahead forecasts.

At this stage it would be useful to state that MAPE is calculated as follows:

$$\text{MAPE} = \frac{1}{k} \sum_{i=1}^k \left| \frac{\hat{x}_i - x_i}{x_i} \right| \times 100\%$$

Table 1

Mean absolute percentage errors (MAPE) of the day-ahead for every estimation method for the whole year 2006. Best results are emphasized in bold.

MAPE	Estimation technique
1.87%	MMPF
2.35%	AIC
1.98%	AICC
2.11%	BIC

Table 2

Recorded and forecast period.

Case	Recorded period	Forecast period
1	5/1/2004–5/3/2004	8/3/2004–14/3/2004 Monday–Sunday (Spring)
2	5/4/2004–5/6/2004	7/6/2004–13/6/2004 Monday–Sunday (Summer)
3	26/7/2004–10/9/2004	13/9/2004–19/09/2004 Monday–Sunday (Autumn)
4	25/10/2004–03/12/2004	6/12/2004–12/12/2004 Monday–Sunday (Winter)

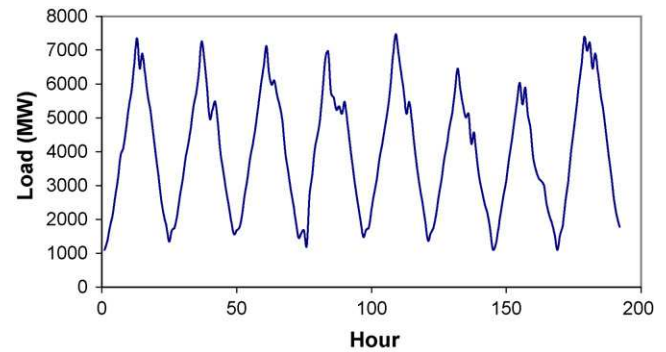


Fig. 14. Typical weekly load demand (January 2004, Monday 12th–Monday 19th).

3.2. Week ahead load forecast

Until now the assumption made is that the load data is represented by a Gaussian distribution. However this cannot be considered as an arbitrary conclusion. Fig. 14 below shows a typical hourly load curve over a period of eight days (Monday 12/01/2004 to Monday 19/01/2004). It can be seen that the load behavior of weekdays (Mondays to Fridays) and weekends (Saturdays and Sundays) is similar and the difference is the actual peak value of the demand load. So generally speaking the whole load can be considered as Gaussian.

In order to access further the proposed method a one week-ahead forecast is performed. The comparison will be made amongst AIC, AICC, BIC, MMPF and an Artificial Neural Network (ANN), with 20 hidden neurons, eight input neurons and one output neuron. The latter technique is been used for shake of completeness, since the week-ahead prediction is undoubtedly a more demanding task. Additionally the forecast of the daily peak is also investigated. Table 2 shows the data used for the criteria comparison (recorded period) and the target forecast period. The procedure is as follows. A period of about two months is used (recorded period) and a prediction for the weak-ahead is made by using the four mentioned criteria. The results are presented in Tables 3–6.

Tables 3–6 indicate that the model performs well with all criteria as far as the one week-ahead prediction and daily peak prediction is concerned. To be more specific the current industry standard for relative error, $(\text{Actual Value} - \text{Forecasted Value})/\text{Actual Value} \times 100\%$, is a value smaller than 2.5% and as it can be seen all the criteria achieve this. Also from the same tables is obvious that the proposed algorithm along with ANN exhibit the smaller average errors.

However we can see that for weekends (Saturdays and Sundays) the relative error is much larger. This is most probably due to the fact that the load during weekends exhibits a non-Gaussian behavior. For these cases AICC (13/3, 19/9, 21/11) and ANN (14/3, 12-13/6, 18/9, 21/11) achieve the better performance. A solution to this problem of non-Gaussian data can be the application of the combination of MMPF with Genetic Algorithms that produces a hybrid algorithm that is shown to overcome such difficulties. This is left however for further research.

Table 3

Case 1: March 8th–March 14th. Average relative forecast error and forecasted daily peak load.

Day, actual peak load (MW)	Criteria forecasted peak load (MW) (relative error (%))				
	MMPF	AICC	AIC	BIC	ANN
Monday 8th, 7345	7257 (1.21)	7244 (1.37)	7176 (2.3)	7208 (1.87)	7203 (1.93)
Tuesday 9th, 7254	7158 (1.32)	7224 (0.42)	7225 (0.39)	7235 (0.26)	7212 (0.58)
Wednesday 10th, 7123	7103 (0.27)	7077 (0.65)	7016 (1.5)	7021 (1.43)	7029 (1.32)
Thursday 11th, 7018	6996 (0.31)	6991 (0.38)	6982 (0.51)	6983 (0.49)	6958 (0.86)
Friday 12th, 7869	7837 (0.41)	7686 (2.33)	7685 (2.34)	7713 (1.98)	7816 (0.67)
Saturday 13th, 6672	6325 (5.2)	6425 (3.7)	6384 (4.32)	6404 (4.01)	6415 (3.85)
Sunday 14th, 6125	5970 (4.5)	5938 (5.01)	5926 (5.2)	5937 (5.03)	5995 (4.1)
Average predicted error (%)	1.89	1.98	2.36	2.15	1.90

Table 4

Case 2: June 7th–June 13th. Average relative forecast error and forecasted daily peak load.

Day, actual peak load (MW)	Criteria forecasted peak load (MW) (relative error (%))				
	MMPF	AICC	AIC	BIC	ANN
Monday 7th, 8456	8421 (0.41)	8420 (0.43)	8253 (2.4)	8316 (1.65)	7203 (1.93)
Tuesday 8th, 8219	8193 (0.32)	8162 (0.69)	8175 (0.53)	8175 (0.53)	7212 (0.58)
Wednesday 9th, 8861	8744 (1.32)	8720 (1.59)	8686 (1.98)	8675 (2.1)	7029 (1.32)
Thursday 10th, 8533	8490 (0.531)	8472 (0.72)	8472 (0.72)	8493 (0.47)	6958 (0.86)
Friday 11th, 8653	8549 (1.2)	8506 (1.7)	8549 (1.2)	8569 (0.97)	7419 (0.67)
Saturday 12th, 8126	7728 (4.9)	7718 (5)	7711 (5.1)	7718 (5.02)	6415 (3.85)
Sunday 13th, 8087	7707 (4.7)	7715 (4.6)	7690 (4.9)	7696 (4.83)	5995 (4.1)
Average predicted error (%)	1.91	2.10	2.40	2.22	1.90

Table 5

Case 3: September 13th–September 19th. Average relative forecast error and forecasted daily peak load.

Day, actual peak load (MW)	Criteria forecasted peak load (mw) (relative error (%))				
	MMPF	AICC	AIC	BIC	ANN
Monday 13th, 9845	9727 (1.2)	9775 (0.71)	9716 (1.31)	9781 (0.65)	9661 (1.87)
Tuesday 14th, 9743	9708 (0.36)	9705 (0.39)	9705 (0.39)	9572 (1.75)	9683 (0.62)
Wednesday 15th, 9821	9688 (1.35)	9683 (1.41)	9607 (2.18)	9744 (0.78)	9697 (1.26)
Thursday 16th, 9654	9616 (0.39)	9565 (0.92)	9565 (0.92)	9607 (0.49)	9584 (0.73)
Friday 17th, 9549	9476 (0.76)	9348 (2.1)	9354 (2.04)	9455 (0.98)	9493 (0.59)
Saturday 18th, 9323	8905 (4.48)	8903 (4.5)	8903 (4.5)	8838 (5.2)	8949 (4.01)
Sunday 19th, 9256	8828 (4.62)	8895 (3.9)	8775 (5.2)	8801 (4.92)	8839 (4.5)
Average predicted error (%)	1.88	1.99	2.36	2.11	1.94

Table 6

Case 4: December 6th–December 12th. Average relative forecast error and forecasted daily peak load.

Day, actual peak load (MW)	Criteria forecasted peak load (mw) (relative error (%))				
	MMPF	AICC	AIC	BIC	ANN
Monday 6th, 9421	9374 (1.2)	9323 (1.04)	9318 (1.31)	9223 (2.1)	9234 (1.98)
Tuesday 7th, 9512	9410 (0.36)	9410 (1.07)	9427 (0.39)	9438 (0.78)	9462 (0.53)
Wednesday 8th, 9318	9269 (1.35)	9219 (1.06)	9090 (2.18)	9220 (1.05)	9229 (0.95)
Thursday 9th, 9224	9162 (0.39)	9149 (0.81)	9154 (0.92)	9154 (0.76)	9154 (0.76)
Friday 10th, 9471	9342 (0.76)	9315 (1.65)	9315 (2.04)	9444 (0.28)	9355 (1.23)
Saturday 11th, 9176	8781 (4.48)	8781 (4.3)	8726 (4.5)	8711 (5.07)	8881 (3.21)
Sunday 12th, 9021	8588 (4.62)	8660 (4.0)	8588 (5.2)	8593 (4.74)	8593 (4.74)
Average predicted error (%)	1.89	1.99	2.36	2.11	1.91

4. Conclusions

Mid-term (i.e. monthly and yearly) electricity load demand forecasting in power systems is quite a complicated task because it is affected by various factors (multiple seasonality, calendar effect, high volatility, etc.). In this work the actual/collected data provided by Hellenic Public Power Corporation S.A., was appropriately manipulated in order to exclude any seasonalities for the period from January 1st 2004 to December 31st 2005. Then an AutoRegressive Moving Average (ARMA) model is fitted (off-line) on this data using the Corrected Akaike Information Criterion (AICC). The

developed model is shown to fit the data in a successful manner. Consequently this model was used by four different estimation methods, in an adaptive day-ahead prediction, in order to predict the electric load for the period from January 1st 2006 to December 31st 2006. All methods performed satisfactorily as far as the load prediction is concerned; however MMPF was able to achieve a slightly better performance in terms of Mean Absolute Percentage Error. This work can be useful in the studies that concern electricity consumption and electricity prices forecasts giving the possibility to the electricity providers, retailers and regulatory authorities to supply uninterrupted energy at a low cost.

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References

- [1] H. Kararli, Further improvement of temporal resolution of seismic data by autoregressive (AR) spectral extrapolation, *Journal of Applied Geophysics* 59 (4) (2006) 324–336.
- [2] G. Madden, J. Tan, Forecasting telecommunications data with linear models, *Telecommunications Policy* 31 (1) (2007) 31–44.
- [3] H.K. Ozturk, H. Ceylan, O.E. Canyurt, A. Hepbasli, Electricity estimation using genetic algorithm approach: a case study of Turkey, *Energy* 30 (2005) 1003–1012.
- [4] B. Kermanshahi, H. Iwamiya, Up to the year 2020 load forecasting using neural nets, *Electric Power Energy Systems* 17 (2002) 789–797.
- [5] M. Yang, X. Yu, China's rural electricity market—a quantitative analysis, *Energy* 29 (2004) 961–977.
- [6] Shyh-ljer Huang, Kuang-Rong Shih, Short-term load forecasting via ARMA model identification including non-Gaussian process considerations, *IEEE Transaction on Power Systems* 18 (2) (2003) 673–679.
- [7] M. Espinoza, C. Joye, R. Belmans, B. DeMoor, Short-term load forecasting, profile identification, and customer segmentation: a methodology based on periodic time series, *IEEE Transaction on Power Systems* 20 (23) (2005) 1622–1630.
- [8] C.C. Chen, R.A. Davis, P.J. Brockwell, Order determination for multivariate autoregressive processes using resampling methods, *Journal of Multivariate Analysis* 57 (1996) 175–190.
- [9] G. Schwarz, Estimation of the dimension of the model, *Annals of Statistics* 6 (1978) 461–464.
- [10] H. Akaike, Fitting autoregressive models for prediction, *Annals of the Institute of Statistical Mathematics* 21 (1969) 243–247.
- [11] PPC S.A., Annual Electrical Energy's Statistical and Economical Data, Hellenic Public Power Corporation S.A., Athens, 2006.
- [12] J. Nowicka-Zagrajek, R. Weron, Modeling electricity loads in California: ARMA models with hyperbolic noise, *Signal Processing* 82 (12) (2002) 1903–1915.
- [13] R. Weron, B. Kozłowska, J. Nowicka-Zagrajek, Modeling electricity loads in California: a continuous-time approach, *Physica A: Statistical Mechanics and Its Applications* 299 (1–2) (2001) 344–350.
- [14] <http://www.im.pwr.wroc.pl/~rweron/MFE.html>.
- [15] J. Rissanen, Modeling by shortest data description, *Automatica* 14 (1978) 465–471.
- [16] J. Rissanen, A predictive least squares principle, *IMA Journal of Mathematical Control and Information* 3 (1986) 211–222.
- [17] M. Wax, Order selection for AR models by predictive least squares, *IEEE Transaction on Acoustics Speech and Signal Processing* 36 (1988) 581–588.
- [18] A.P. Sage, G.W. Husa, Adaptive filtering with unknown prior statistics, in: *Proceedings of Joint Automatic Control Conference*, Boulder, Colorado, 1969, pp. 760–769.
- [19] D.G. Lainiotis, Optimal adaptive estimation: structure and parameter adaptation, *IEEE Transaction on Automatic Control* AC-16 (1971) 160–170.
- [20] D.G. Lainiotis, Partitioning: a unifying framework for adaptive systems I: estimation, *Proceedings of the IEEE* 64 (8) (1976) 1126–1143.
- [21] D.G. Lainiotis, Partitioning: a unifying framework for adaptive systems II: control, *Proceedings of the IEEE* 64 (8) (1976) 1182–1198.
- [22] K. Watanabe, *Adaptive Estimation and Control: Partitioning Approach*, Prentice Hall, Englewood Cliffs, NJ, 1992.
- [23] S.Sp. Pappas, A.K. Leros, S.K. Katsikas, Joint order and parameter estimation of multivariate autoregressive models using multi-model partitioning theory, *Digital Signal Processing* 16 (6) (2006) 782–795.
- [24] S.K. Katsikas, S.D. Likothanassis, G.N. Beligiannis, K.G. Berkeris, D.A. Fotakis, Genetically determined variable structure multiple model estimation, *IEEE Transaction on Signal Processing* 49 (10) (2001) 2253–2261.
- [25] D.G. Lainiotis, P. Papaparaskeva, A partitioned adaptive approach to nonlinear channel equalization, *IEEE Transaction on Communications* 46 (10) (1998) 1325–1336.
- [26] V.C. Moussas, S.D. Likothanassis, S.K. Katsikas, A.K. Leros, Adaptive on-line multiple source detection, in: *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP'05)*, vol. 4, 2005, pp. 1029–1032.
- [27] D.G. Lainiotis, S.K. Katsikas, S.D. Likothanassis, Adaptive deconvolution of seismic signals: performance, computational analysis, parallelism, *IEEE Transaction on Acoustics, Speech, and Signal Processing* 36 (11) (1988) 1715–1734.
- [28] N.V. Nikitakos, A.K. Leros, S.K. Katsikas, Towed array shape estimation using multimodel partitioning filters, *IEEE Journal of Oceanic Engineering* 23 (4) (1998) 380–384.
- [29] V.C. Moussas, S.Sp. Pappas, Adaptive network anomaly detection using bandwidth utilization data, in: *1st International Conference on Experiments/Processes/System Modelling/Simulation/Optimization*, Athens, Greece, 2005.
- [30] V.C. Moussas, S.K. Katsikas, A multi-model approach to fatigue crack growth monitoring and prediction, in: *12th International Workshop on Systems, Signals & Image Processing*, Chalkida, Greece, 2005, pp. 57–61.
- [31] R.L. Sengbush, D.G. Lainiotis, Simplified parameter quantization procedure for adaptive estimation, *IEEE Transaction on Automatic Control* AC-14 (1969) 424–425.
- [32] B.D.O. Anderson, T.S. Brinsmead, F. De Bruyne, J. Hespanha, D. Liberzon, A.S. Morse, Multiple model adaptive control, Part 1: finite controller coverings, *International Journal of Robust and Nonlinear Control* 10 (2000) 909–929.
- [33] R.M. Hawks, J.B. Moore, Performance of Bayesian parameter estimators for linear signal models, *IEEE Transaction on Automatic Control* AC-21 (1976) 523–527.
- [34] G. Beligiannis, L. Skarlas, S. Likothanassis, A generic applied evolutionary hybrid technique, *IEEE Signal Processing Magazine* 21 (3) (2004) 28–38.
- [35] H. Lutkepohl, Comparison of criteria for estimating the order of a vector AR process, *Journal of Time Series Analysis* 6 (1985) 35–52.

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